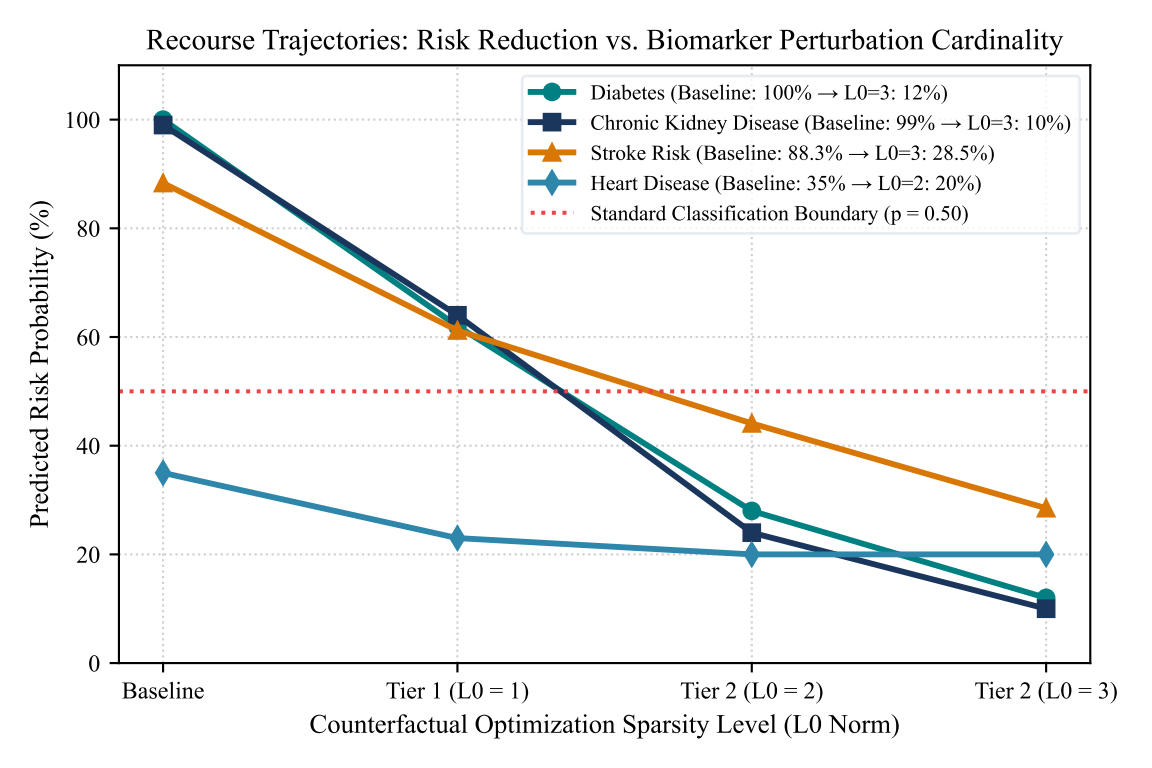
Recourse Trajectories: Risk Reduction vs. Biomarker Perturbation Cardinality
Diabetes (Baseline: 100% → L0=3: 12%)
Chronic Kidney Disease (Baseline: 99% → L0=3: 10%)
Stroke Risk (Baseline: 88.3% → L0=3: 28.5%)
Heart Disease (Baseline: 35% → L0=2: 20%)
Standard Classification Boundary (p = 0.50)
Predicted Risk Probability (%)
0
20
40
60
80
100
Baseline
Tier 1 (L0 = 1)
Tier 2 (L0 = 2)
Tier 2 (L0 = 3)
Counterfactual Optimization Sparsity Level (L0 Norm)